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IMI

Optimized selection of marker genes for spatial transcriptomics deconvolution

Introduction

Spatial transcriptomics (ST) allows the profiling of gene expression while preserving the tissue structure and therefore the cells' location.

Common approaches lack single cell resolution. Therefore, a main analysis of ST data is to infer cell type composition of each spot, also known as **deconvolution.**

Most of the tools available for deconvolution require
- (Matching) single cell dataset
- Set of **marker genes** defining each cell type

Usually defined via the top differentially expressed genes

⚠ Considers changes in expression levels and their significance
DOES NOT consider specificity of the expression

Methods

We performed a detailed analysis of genes and their potential eligibility as markers, considering changes in **expression level and specificity of gene expression**. Based on a single cell dataset and spatial transcriptomics data from six cases of hepatoblastoma, we compared different approaches for filtering differentially expressed genes and evaluated the results based on the specificity and strength of the signature plotted on single cell UMAP plots and the deconvolution results.

Results

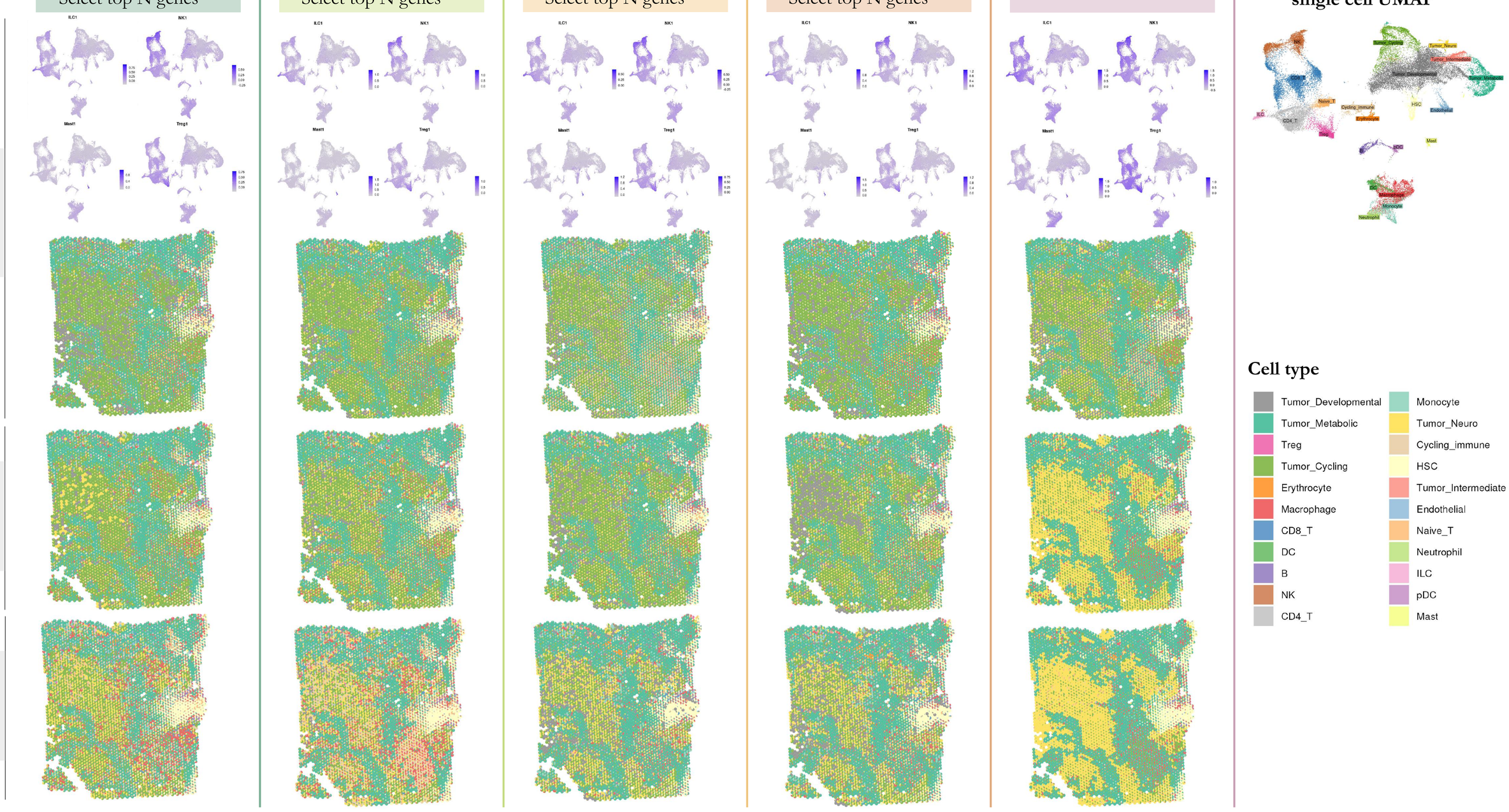
Approach 1:
Delta in top 25%
Pct.2 in bottom 25%
Sort by Log2FC
Select top N genes

Approach 2:
Delta in top 10%
Pct.2 in bottom 25%
Sort by Log2FC
Select top N genes

Approach 3:
Log2FC in top 25%
Pct.2 in bottom 25%
Sort by delta
Select top N genes

Approach 4:
Log2FC in top 10%
Pct.2 in bottom 25%
Sort by delta
Select top N genes

Approach 5:
Sort by Log2FC
Select top N genes



Conclusions

Filtering approaches that consider the specificity as well as the strength of the gene expression are shown to result in a **more accurate** and **stable** deconvolution of the spatial data in comparison with the ones that only consider the strength of gene expression.

References

UKM Universitätsklinikum Münster

WWU MÜNSTER

medizinische fakultät Westfälische Wilhelms-Universität Münster